

Let's work through a **complete XGBoost classification example** and see how it selects the root node. Unlike regression, XGBoost classification uses **logistic loss**, so the gradients and Hessians are different.

---

### Step 1: Dataset

Binary classification:

| Sample | x | Class y |
|--------|---|---------|
| 1      | 1 | 0       |
| 2      | 2 | 0       |
| 3      | 3 | 0       |
| 4      | 4 | 1       |
| 5      | 5 | 1       |
| 6      | 6 | 1       |

Goal:

$$y \in \{0,1\}$$

---

### Step 2: Initial Prediction

For binary classification, XGBoost works with **log-odds scores**.

Initial score:

$$F_0 = 0$$

Probability:

$$p = \frac{1}{1 + e^{-F_0}}$$

Since  $F_0 = 0$ ,

$$p = 0.5$$

Every sample initially has probability 0.5.

---

### Step 3: Logistic Loss

Binary cross-entropy:

$$L = -y \log(p) - (1 - y) \log(1 - p)$$

---

### Step 4: Gradients and Hessians

For XGBoost classification:

Gradient:

$$g_i = p_i - y_i$$

Hessian:

$$h_i = p_i(1 - p_i)$$

At initialization:

$$p_i = 0.5$$

Therefore

$$h_i = 0.25$$

for every sample.

---

### Compute Gradients

For class 0:

$$g = 0.5 - 0 = 0.5$$

For class 1:

$$g = 0.5 - 1 = -0.5$$

---

| Sample | y | p   | g    | h    |
|--------|---|-----|------|------|
| 1      | 0 | 0.5 | 0.5  | 0.25 |
| 2      | 0 | 0.5 | 0.5  | 0.25 |
| 3      | 0 | 0.5 | 0.5  | 0.25 |
| 4      | 1 | 0.5 | -0.5 | 0.25 |
| 5      | 1 | 0.5 | -0.5 | 0.25 |
| 6      | 1 | 0.5 | -0.5 | 0.25 |

---

#### Step 5: Root Node

Total gradient:

$$G = 0.5 + 0.5 + 0.5 - 0.5 - 0.5 - 0.5$$
$$G = 0$$

Total Hessian:

$$H = 6 \times 0.25$$
$$H = 1.5$$

Assume

$$\lambda = 1$$

---

#### Step 6: Evaluate Candidate Splits

Possible splits:

$$x < 1.5$$

$$x < 2.5$$

$$x < 3.5$$

$$x < 4.5$$

$$x < 5.5$$

Gain formula:

$$Gain = \frac{1}{2} \left( \frac{G_L^2}{H_L + \lambda} + \frac{G_R^2}{H_R + \lambda} - \frac{G^2}{H + \lambda} \right)$$

Since  $G = 0$ , the last term is zero.

---

**Candidate 1:  $x < 1.5$**

Left:

Sample 1

$$G_L = 0.5$$

$$H_L = 0.25$$

Right:

$$G_R = -0.5$$

$$H_R = 1.25$$

Gain:

$$\begin{aligned} & \frac{1}{2} \left( \frac{0.25}{1.25} + \frac{0.25}{2.25} \right) \\ & = 0.1556 \end{aligned}$$

---

**Candidate 2:  $x < 2.5$**

Left:

Samples 1,2

$$G_L = 1.0$$

$$H_L = 0.5$$

Right:

$$G_R = -1.0$$

$$H_R = 1.0$$

Gain:

$$\frac{1}{2} \left( \frac{1}{1.5} + \frac{1}{2} \right)$$

$$= 0.5833$$

---

**Candidate 3:  $x < 3.5$**

Left:

Samples 1,2,3

$$G_L = 1.5$$

$$H_L = 0.75$$

Right:

Samples 4,5,6

$$G_R = -1.5$$

$$H_R = 0.75$$

Gain:

$$\frac{1}{2} \left( \frac{2.25}{1.75} + \frac{2.25}{1.75} \right)$$

$$= 1.2857$$

---

**Candidate 4:  $x < 4.5$**

Gain:

0.5833

---

**Candidate 5:  $x < 5.5$**

Gain:

0.1556

---

**Root Node Selection**

**Split Gain**

$x < 1.5$  0.1556

$x < 2.5$  0.5833

$x < 3.5$  1.2857

$x < 4.5$  0.5833

$x < 5.5$  0.1556

Highest gain:

1.2857

Therefore XGBoost chooses

$x < 3.5$

as the root split.

---

**Step 7: Compute Leaf Scores**

Leaf score formula:

$$w = -\frac{G}{H + \lambda}$$

---

**Left Leaf**

$$\begin{aligned} G &= 1.5 \\ H &= 0.75 \\ w_L &= -\frac{1.5}{1.75} \\ &= -0.857 \end{aligned}$$

---

**Right Leaf**

$$\begin{aligned} G &= -1.5 \\ H &= 0.75 \\ w_R &= 0.857 \end{aligned}$$

---

### **Step 8: Update Predictions**

Initial score:

$$F_0 = 0$$

New scores:

Left leaf:

$$F = -0.857$$

Right leaf:

$$F = 0.857$$

Convert scores to probabilities:

$$p = \frac{1}{1 + e^{-F}}$$


---

Left side:

$$p = \frac{1}{1 + e^{0.857}}$$

$$p = 0.298$$


---

Right side:

$$p = \frac{1}{1 + e^{-0.857}}$$

$$p = 0.702$$


---

### Final Classification After First Tree

**x Probability(Class=1)**

1 0.298

2 0.298

3 0.298

4 0.702

5 0.702

6 0.702

Using threshold 0.5:

**x Predicted Class**

1 0



### **x Predicted Class**

2 0

3 0

4 1

5 1

6 1

All samples are correctly classified after just one tree.

---

### **Key Difference from Regression**

For **regression**:

$$\begin{aligned}g &= \hat{y} - y \\h &= 1\end{aligned}$$

For **binary classification**:

$$\begin{aligned}g &= p - y \\h &= p(1 - p)\end{aligned}$$

Everything else—the gain formula, leaf-weight formula, and greedy split selection—is exactly the same. The only difference is how gradients and Hessians are computed from the loss function.